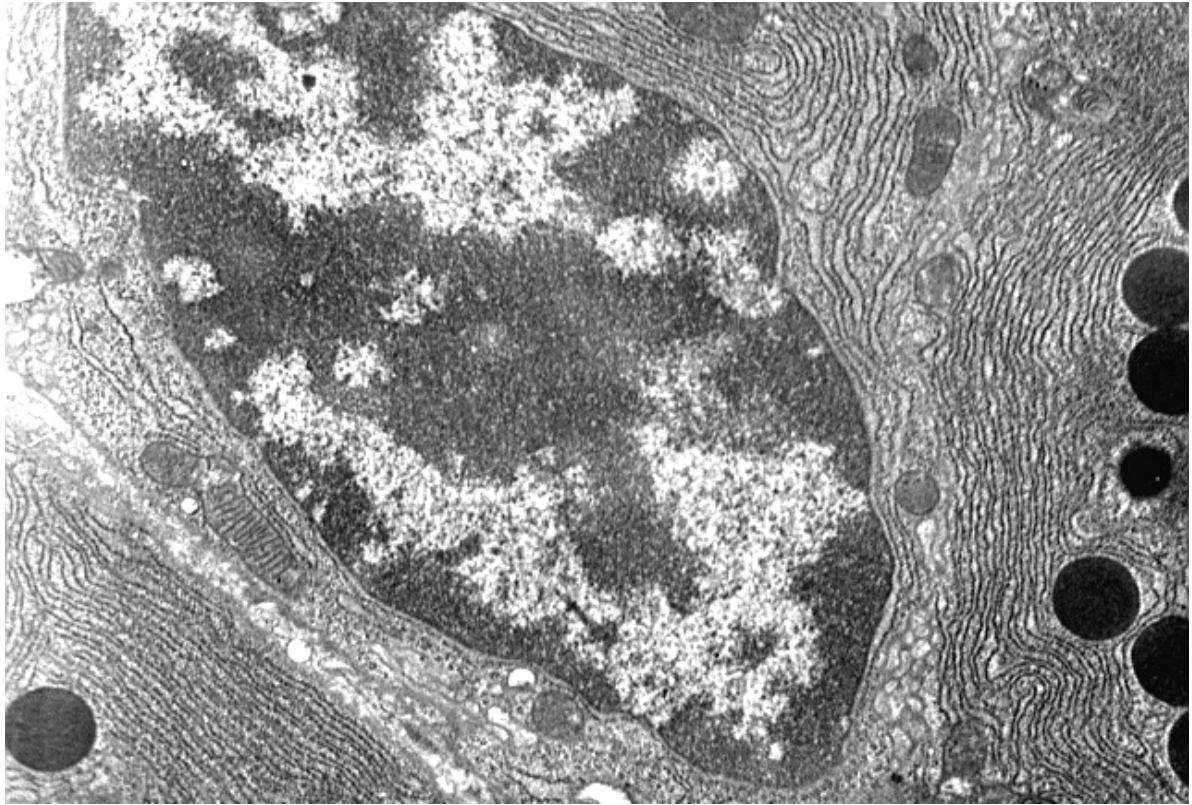


- 1 The electron micrograph shows a cell from the pancreas.

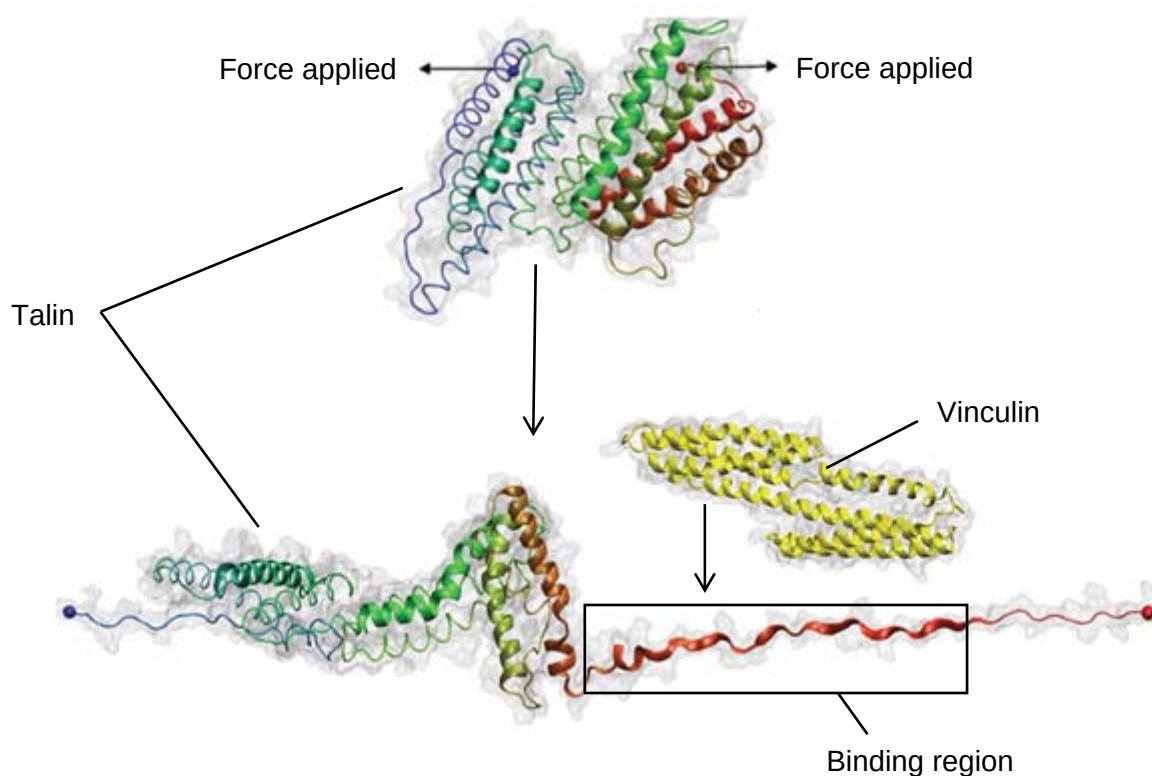


Which of the following are some visible structural features of the cell that enable it to function in secretion?

I	Presence of mitochondria to provide energy in the form of ATP for the process of secretion.
II	Presence of Golgi apparatus for the packaging and modification of proteins.
III	Presence of bound ribosomes for the synthesis of secretory proteins.
IV	Presence of numerous lysosomes for the digestion of macromolecules.
V	Presence of peroxisomes to aid in the process of transport of proteins to the cell surface membrane.

- A I and III only
 B II and IV only
 C I, II and III only
 D All of the above

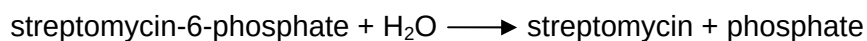
- 2 The diagram shows how the 3D conformation of the protein talin can be altered to expose a binding region for another protein, vinculin, to bind.



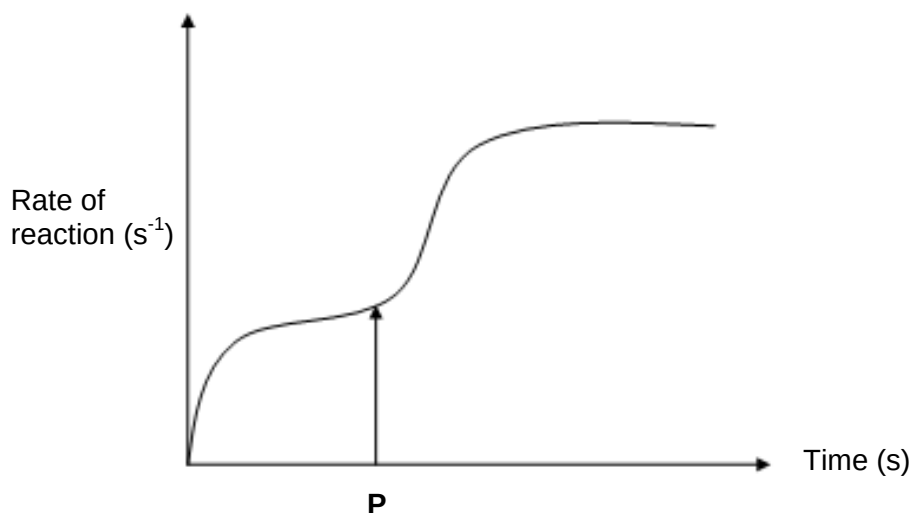
Which of the following changes must occur to the structure of talin to allow vinculin to bind to it when stretched?

	Primary structure	Secondary structure	Tertiary structure
A	No change	Ionic bonds broken	Hydrophobic interactions disrupted
B	No change	Hydrogen bonds broken	Ionic bonds broken
C	No change	Ionic bonds broken	Disulfide bonds disrupted
D	Peptide bonds broken	Hydrogen bonds broken	Ionic bonds broken

- 3 The enzyme streptomycin-6-phosphatase catalyses the final step in the production of the antibiotic streptomycin optimally at 30°C and pH 7.



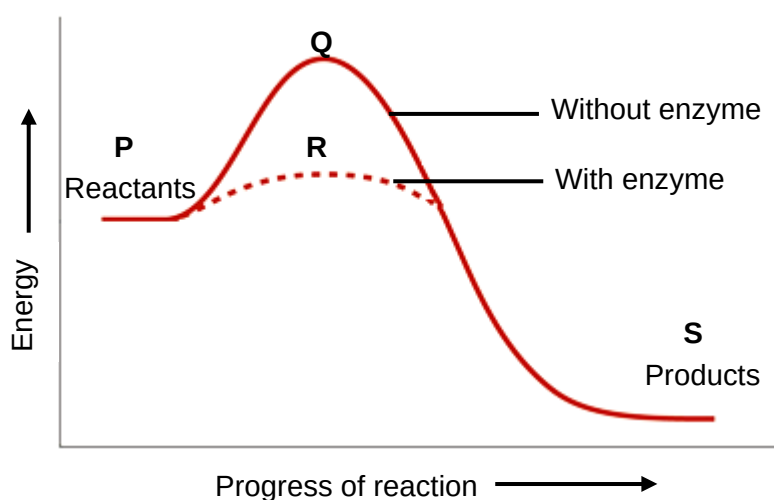
In one experiment, the time course of the reaction was followed by measuring the rate of reaction at regular time intervals, as shown in the graph.



Given that there was a change in one experimental condition at **P**, which of the following cannot be the change introduced?

- A Increasing temperature by 10°C from 15°C.
- B Doubling streptomycin-6-phosphate concentration.
- C Increasing pH from 7 to 10.
- D Doubling streptomycin-6-phosphatase concentration.

- 4 The effect of addition of an enzyme on the energy required for a reaction to proceed is shown in the graph.

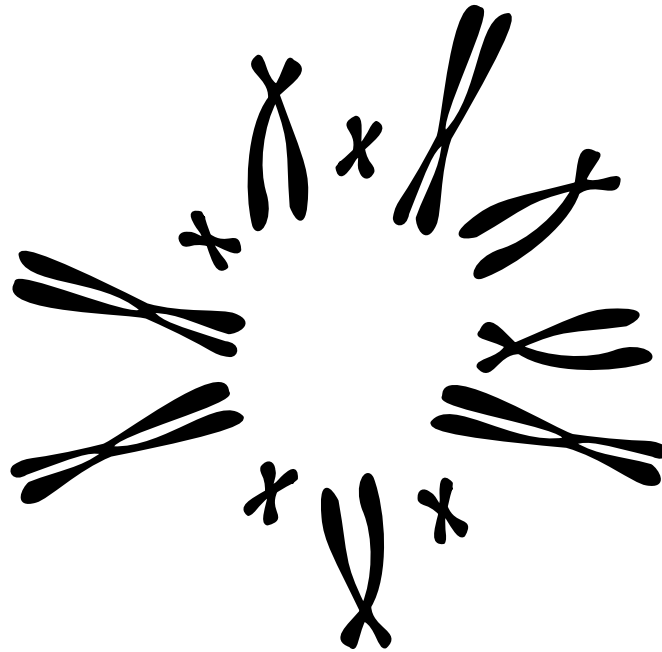


Which of the following statements accurately describe the events occurring at **P**, **Q**, **R** and **S**?

I	Addition of enzymes to a reaction would favour the formation of more products at S than the uncatalysed reaction.
II	Increasing the concentration of reactants at P would not result in a change in the energy values at Q and R .
III	Increasing the temperature of the reaction up to the optimum would not reduce the energy difference between Q and R .
IV	There would be a smaller number of molecules at R than at Q .

- A** I and II only
B II and III only
C III and IV only
D All of the above

- 5 The diagram shows the chromosomal behaviour in an animal cell during a particular stage of cell division.



Which of the following correctly describes the chromosomes shown?

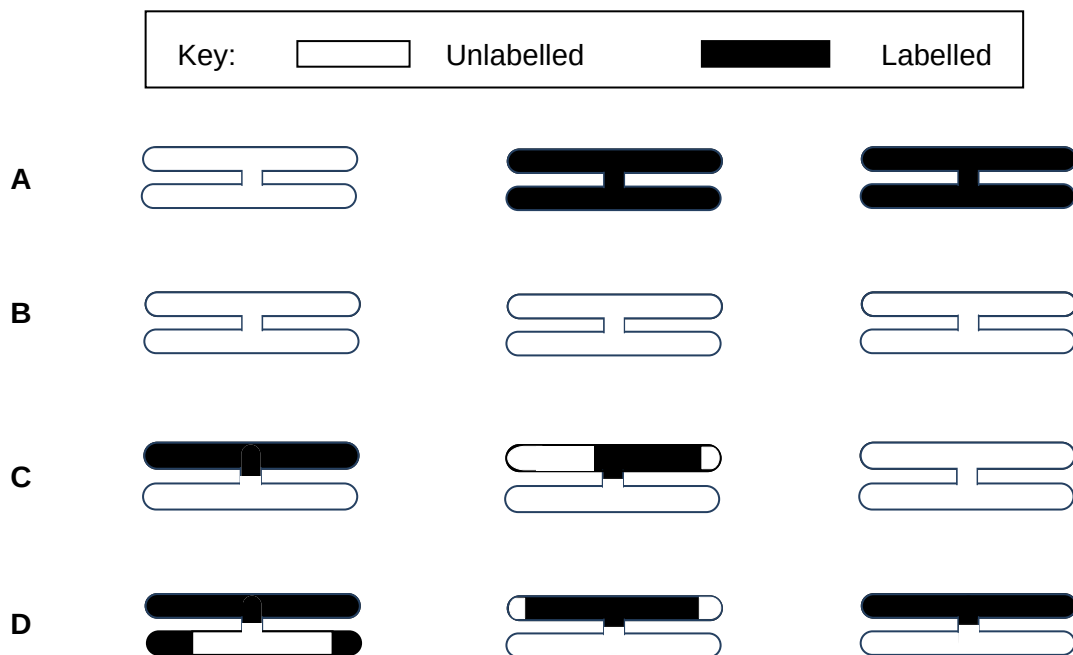
	Stage of cell division	Number of chromosomes in a haploid set	Total number of DNA strands
A	Metaphase	4	48
B	Prophase	2	24
C	Prophase	3	48
D	Metaphase	3	48

- 6 Which of the following is **not** true of adult stem cells during tissue repair?
- A The stem cells must have active telomerase.
 - B The different checkpoints in the cell cycle of the stem cells are activated.
 - C Mitosis of the stem cells is induced without any stimulus.
 - D The stem cells will stop dividing after the damaged cells are replaced.

7 The following experiment was carried out to investigate cell division in *Allium* cells.

- Diploid cells, containing six chromosomes each, were grown in a medium containing radioactive thymine and allowed to divide once by mitosis.
- Cells in early interphase (before S phase) were then transferred to a medium where the available thymine was not radioactive.
- A single cell was immediately isolated and allowed to divide by meiosis. When the daughter cells reached metaphase II, they were fixed and their chromosomes were inspected for radioactivity.

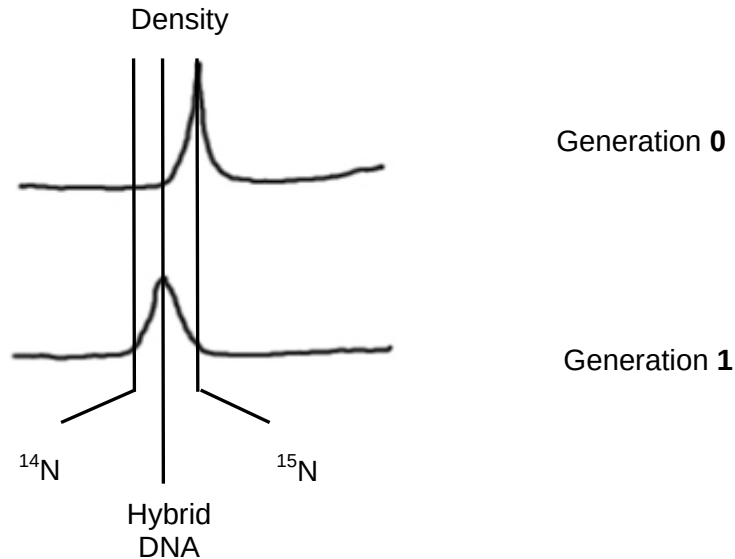
Which of the following diagrams correctly represents the distribution of radioactivity at metaphase II in any daughter cell?



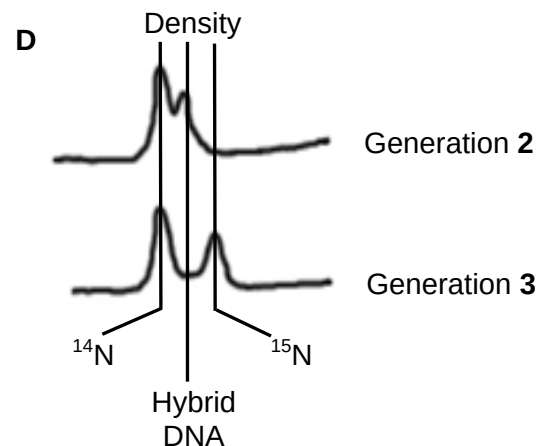
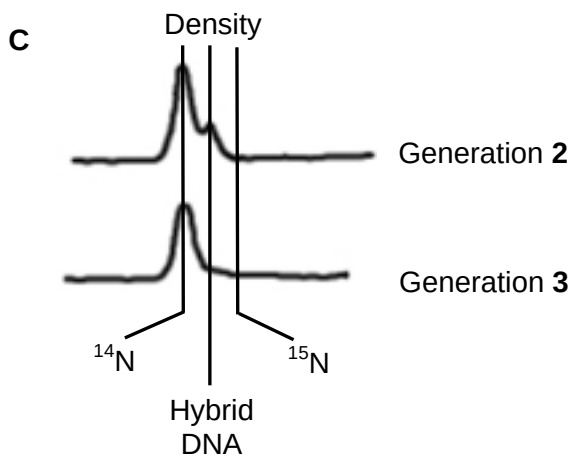
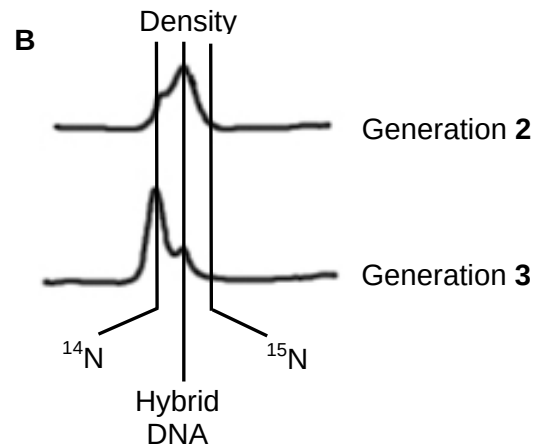
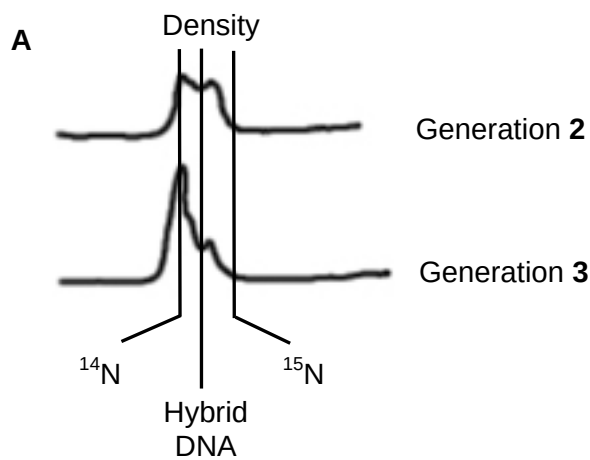
8 A student designs an artificial DNA segment for insertion into the genome of a thermophilic bacterium. Which of the following DNA segments would be the most suitable for insertion?

A	CAACTAGGTT GTTGATCCAA	C	GAACTATCTT CTTGATAGAA
B	CAACTAGGTT GTTGATGCAT	D	GTTCTTGACA CAAGAAGAGT

- 9 The bacterium *Escherichia coli* was grown for many generations in ^{15}N and then subsequently transferred to ^{14}N to grow. Samples of *E. coli* were removed from the ^{14}N culture after each generation. The DNA from each sample was extracted and centrifuged. The diagram shows the results of the experiment for generations 0 and 1. The height of each peak represents the level of radioactivity.



Which of the following diagrams shows the correct densities and relative proportions of the DNA extracted from the *E. coli* at generations 2 and 3?



10 Which of the following statements are true about viruses?

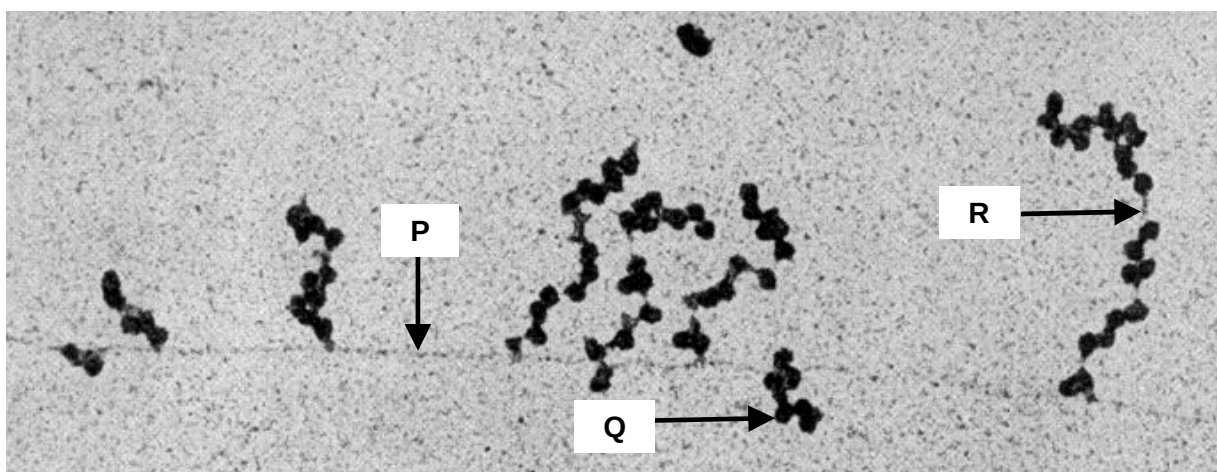
I	They can only replicate within living host cells.
II	The virion is a metabolically inert, complete virus particle in which the viral genome is carried.
III	No single gene is shared by all viruses.
IV	The basic structural components of all viruses include capsid, envelope and nucleic acid core.
V	Viral genomes consist of either DNA and RNA but not both.

- A** I, III and IV only
B III, IV and V only
C I, II, III and V only
D All of the above

11 The genome size of the bacterium *Haemophilus influenzae* is 1830 kb. Bacterial DNA polymerase is found to proceed at a speed of $3.4 \times 10^{-1} \mu\text{ms}^{-1}$. What is the time taken to replicate the genome of *H. influenzae*?

- A** 9 minutes
B 15 minutes
C 30 minutes
D 54 minutes

- 12 The electron micrograph shows three structural components **P**, **Q** and **R** involved in prokaryotic gene expression.

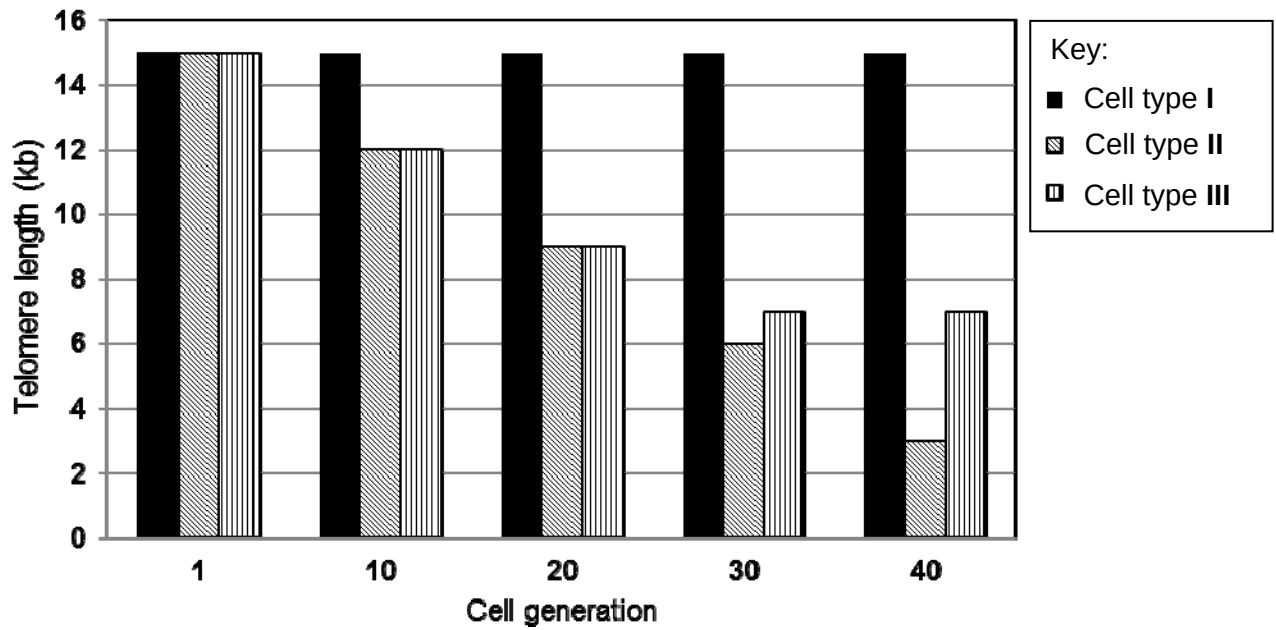


Which of the following statement(s) concerning these three components is / are true?

I	RNA polymerase adds incoming nucleotides to form P .
II	The core enzyme binds to the Pribnow box found in R .
III	16S rRNA in Q recognises and binds to the Shine-Dalgarno sequence.
IV	Q is involved in forming R .

- A III only
 B II and III only
 C I, II and IV only
 D All of the above

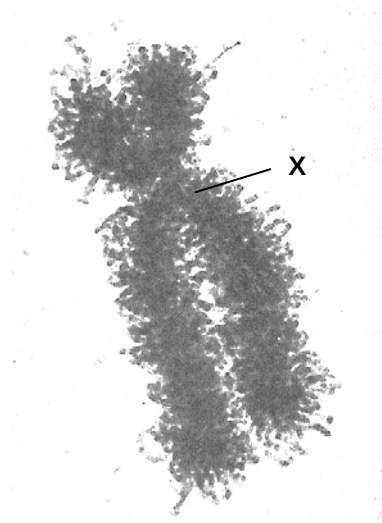
13 The graph shows the telomere length of three different cell types I to III.



Which of the following correctly identifies cell types I to III at generation 40?

	Cell type		
	I	II	III
A	Stem cells	Somatic cells	Tumour cells
B	Stem cells	Tumour cells	Somatic cells
C	Germline cells	Somatic cells	Stem cells
D	Somatic cells	Tumour cells	Germline cells

- 14 The electron micrograph shows a specialised region **X** found on a chromosome.

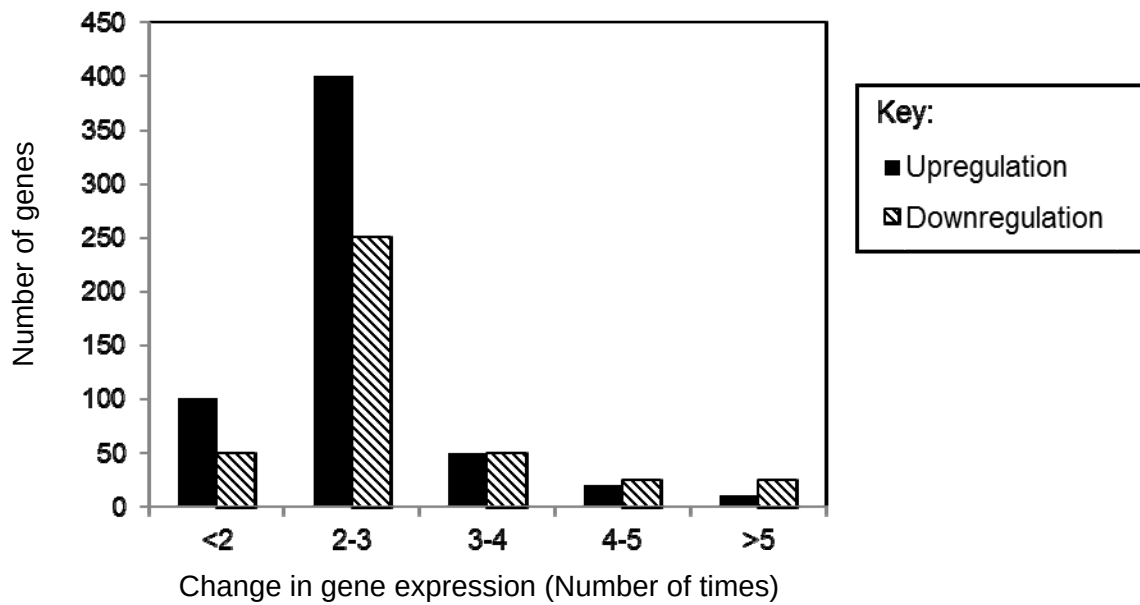


Which of the following statements are true of **X**?

I	With a mutation at X , there is no alignment of chromosomes at metaphase.
II	X is replicated during S phase.
III	X is transcriptionally inactive throughout the cell cycle.
IV	The length of X is highly conserved.

- A** I and IV only
B III and IV only
C I, II and III only
D All of the above

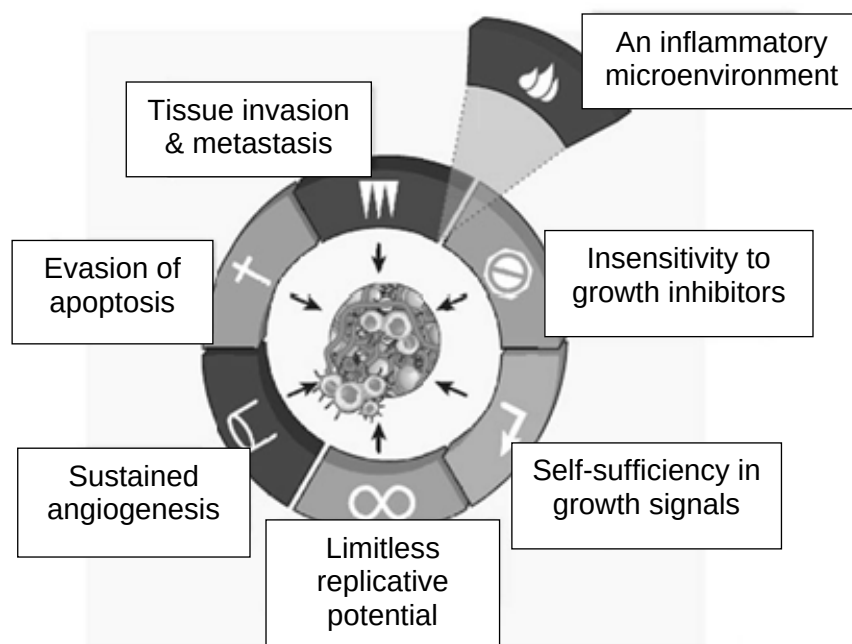
- 15 The graph shows the changes in gene expression in heart cells of a group of mice after oxygen deprivation for four minutes.



Which of the following can result in all the observed changes?

- A Methylation of cytosine at the promoters of the genes.
- B Activation of specific transcription factors.
- C Alternative splicing of mRNA.
- D Increase in activity of histone acetyltransferase.

16 The diagram illustrates the hallmarks of cancer.



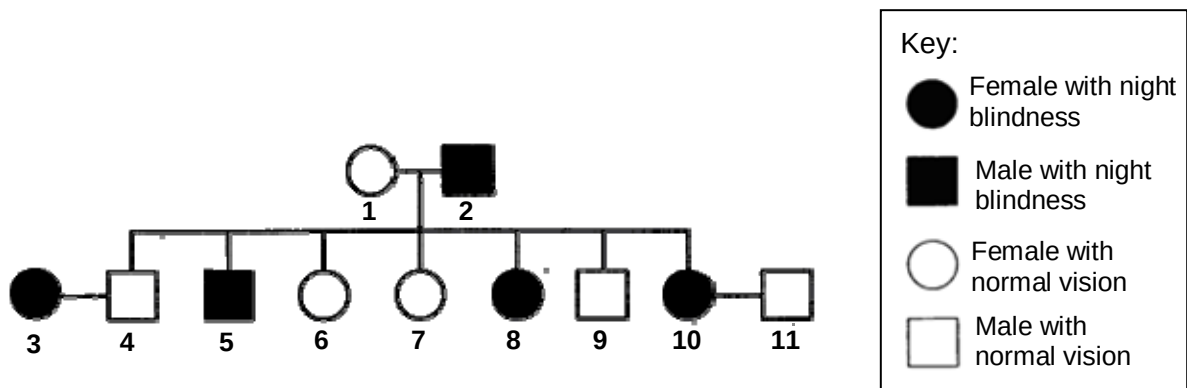
Which of the following statements correctly describe the changes in cancer cells?

I	Limitless replicative potential often results in the accumulation of chromosomal mutations in many cancer cells.
II	Cancer cells could overproduce signal molecules so that they become self-sufficient in growth signals.
III	Angiogenesis is the result of expression of oncogenes in a cell line that produces blood vessels.
IV	Loss-of-function mutations in tumour suppressor genes contribute to tissue invasion and metastasis.

- A I and IV only
- B II and III only
- C I, II and IV only
- D II, III and IV only

- 17** Night blindness is a condition in which affected people have difficulty seeing in dim light. The allele **N** for night blindness is dominant to the allele **n** for normal vision.

The diagram shows part of a family tree illustrating the inheritance of night blindness.



What is the probability that the next two children born to individuals **10** and **11** will be an elder girl and a younger boy both with night blindness?

- A** 0.0625
B 0.125
C 0.25
D 0.50
- 18** The type of plumage (feathers) in chickens results from the expression of an autosomal gene locus that carries two alleles:

Allele **H** (hen-feathering) is dominant in both female and male chickens.

Allele **h** (cock-feathering) is recessive, and the phenotype is expressed only in male chickens.

Which of the following would be the result of a mating between a cock-feathering male chicken, and a hen-feathering female chicken which is heterozygous at the gene locus?

- A** The probability of a male offspring being cock-feathering is 100% and that of a female offspring being hen-feathering is 100%.
B The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 100%.
C The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 50%.
D The probability of a male offspring being hen-feathering is 100% and that of a female offspring being hen-feathering is 100%.

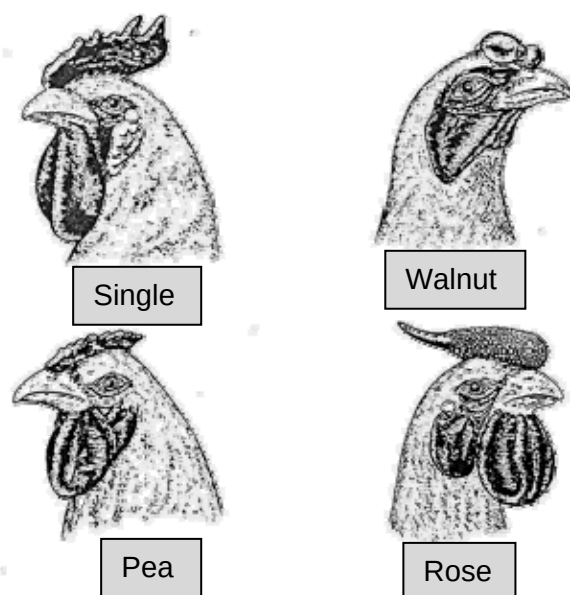
- 19** Pure-breeding tomato plants with red fruit and white flowers were crossed with pure-breeding tomato plants with yellow fruit and yellow flowers. All the F_1 plants had red fruit and yellow flowers. F_1 plants were then crossed with pure-breeding plants with yellow fruit and white flowers. The following results were obtained:

64	red fruit and yellow flowers
333	red fruit and white flowers
350	yellow fruit and yellow flowers
58	yellow fruit and white flowers

Which of the following correctly shows the arrangement of the two genes for fruit colour and flower colour in the F_1 plants and the map distance between them?

	Arrangement of genes	Map distance between genes
A	Coupling	15 cM
B	Coupling	85 cM
C	Repulsion	15 cM
D	Repulsion	85 cM

- 20** In chickens, the pea comb is due to a dominant allele **P** at one gene locus, while the rose comb is due to a dominant allele **R** at another gene locus on a different chromosome. Chickens with both dominant alleles have walnut combs. Chickens without both dominant alleles have single combs.

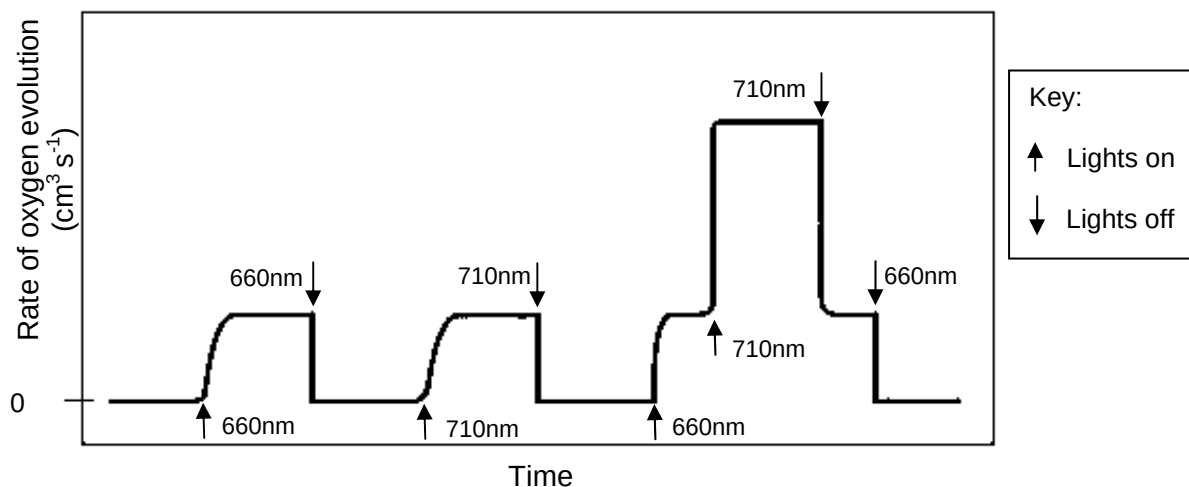


A cross between two chickens produced **14** pea-comb, **15** walnut-comb, **6** rose-comb and **5** single-comb chicks.

What are the genotypes of the parents in this cross?

- A** $rrPp \times Rrpp$
- B** $RrPp \times rrpp$
- C** $RrPp \times Rrpp$
- D** $RrPp \times rrPp$

- 21 Green algae were illuminated with two beams of light of wavelengths 660nm and 710nm, either singly or simultaneously as shown in the graph. The rate of oxygen evolution was subsequently measured.



Which of the following can be concluded from the graph?

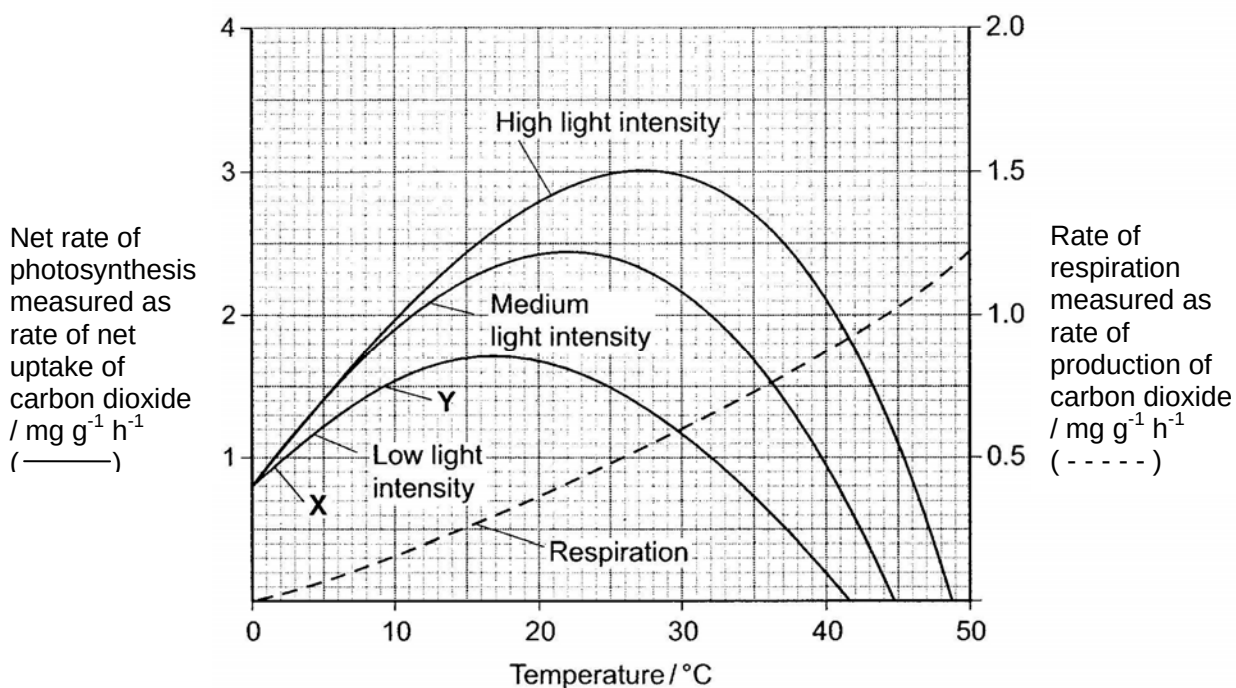
- A Photosynthesis is independent of both light wavelengths of 660nm and 710nm.
 B The rate of photolysis of water is greater at wavelength of 660nm than 710nm.
 C The photosystems of the green algae harvest light most efficiently at an average wavelength of 655nm.
 D Photosynthesis occurs maximally in the presence of both light wavelengths of 660nm and 710nm.
- 22 In a laboratory, a plant with variegated leaves (containing orange and green pigments) was supplied with radioactive carbon dioxide, $^{14}\text{CO}_2$. The plant was kept in the dark for 12 hours and then illuminated for the next 12 hours. A leaf from the plant was obtained and its level of radioactivity was measured both in the absence and presence of light. The results are shown in the table.

	Level of radioactivity in leaf (Arbitrary units)	
	Orange region of leaf	Green region of leaf
Absence of light	225	225
Presence of light	410	9271

Which of the following could be the most likely explanation for the level of radioactivity found in the orange region of the leaf in the presence of light?

- A Some photosynthesis occurs in the orange region but due to the absence of chlorophyll in that region, the rate of photosynthesis is low.
 B Photosynthesis occurs in the orange region but no storage of starch occurs.
 C Photosynthetic products diffuse into the orange region.
 D Radioactive $^{14}\text{CO}_2$ diffuses into the orange region and accumulates in that region.

- 23 Botanists studied the effect of light intensity and temperature on the rates of photosynthesis and respiration in the goldfish plant *Alloplectus nummularia*. The results are shown in the graph.

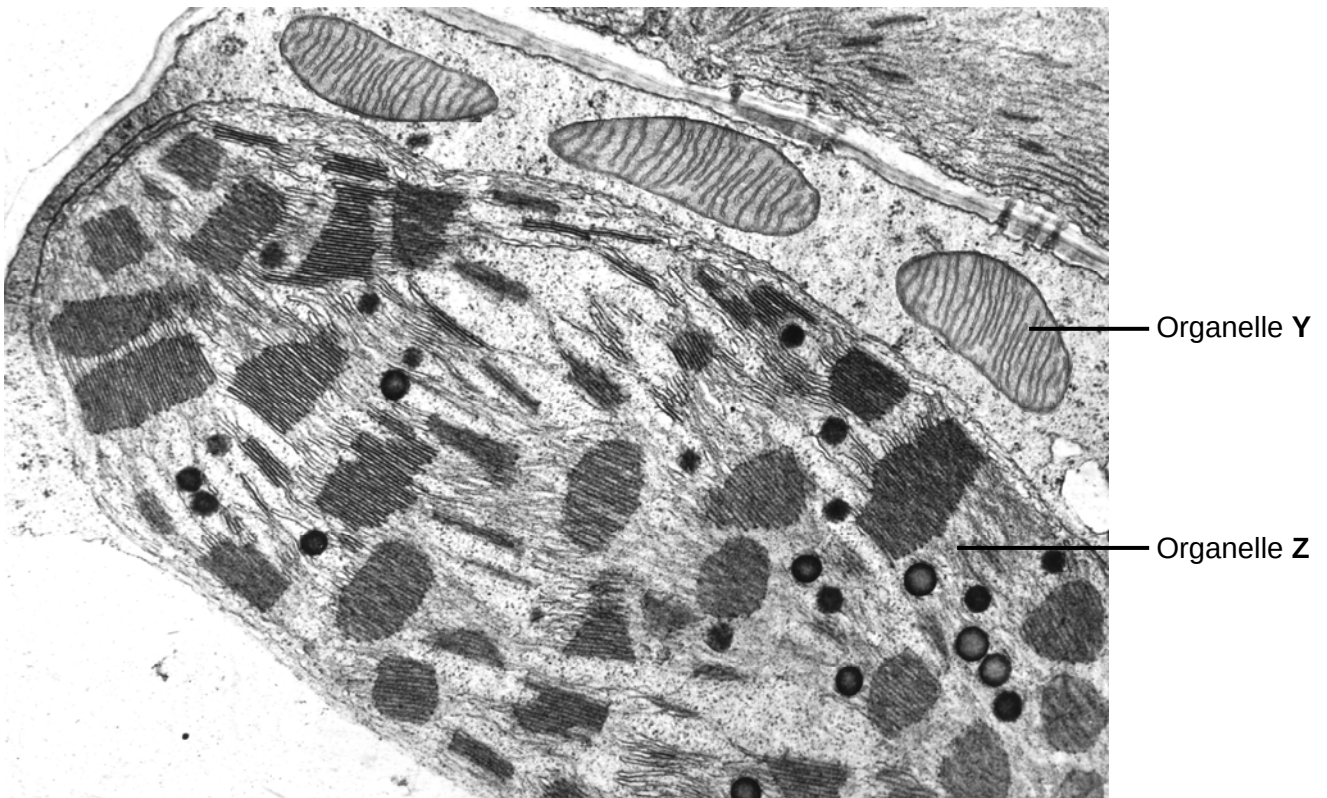


Which of the following statement(s) is / are **not** true?

I	Only light intensity limits the rate of photosynthesis between X and Y.
II	Unlike photosynthesis, the rate of respiration is dependent on temperature.
III	Under low light intensity, the compensation point of <i>A. nummularia</i> occurs at 30°C .
IV	Increase in temperature causes an increase in the rate of production of carbon dioxide by respiration.

- A IV only
- B I and II only
- C I, II and III only
- D All of the above

- 24 A new species of plant was recently discovered in the Amazon forest. The electron micrograph shows two organelles Y and Z in a leaf mesophyll cell of the plant.

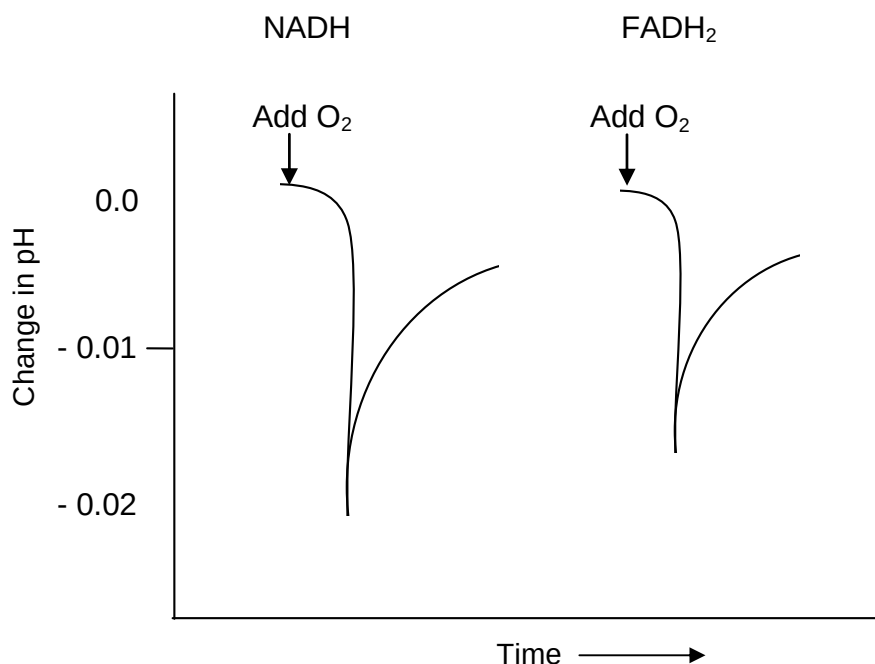


Which of the following statements are **not** true about organelles Y and Z?

I	Organelle Z utilises transporters to export ATP to organelle Y to drive cellular activities.
II	Oxygen released by organelle Z is used in organelle Y during Krebs cycle.
III	Phosphate ions in organelle Y is used for the production of ATP during Calvin cycle.
IV	NADPH molecules produced in organelle Z are used in organelle Y for the production of triose phosphate.

- A I and IV only
 B II and III only
 C II and IV only
 D All of the above

- 25 Isolated mitochondria were incubated with NADH in one experiment and an equal amount of FADH_2 in another experiment. The mitochondria were initially deprived of oxygen. The pH of the intermembrane space was then monitored as a known quantity of oxygen was added. The results are shown in the graph.

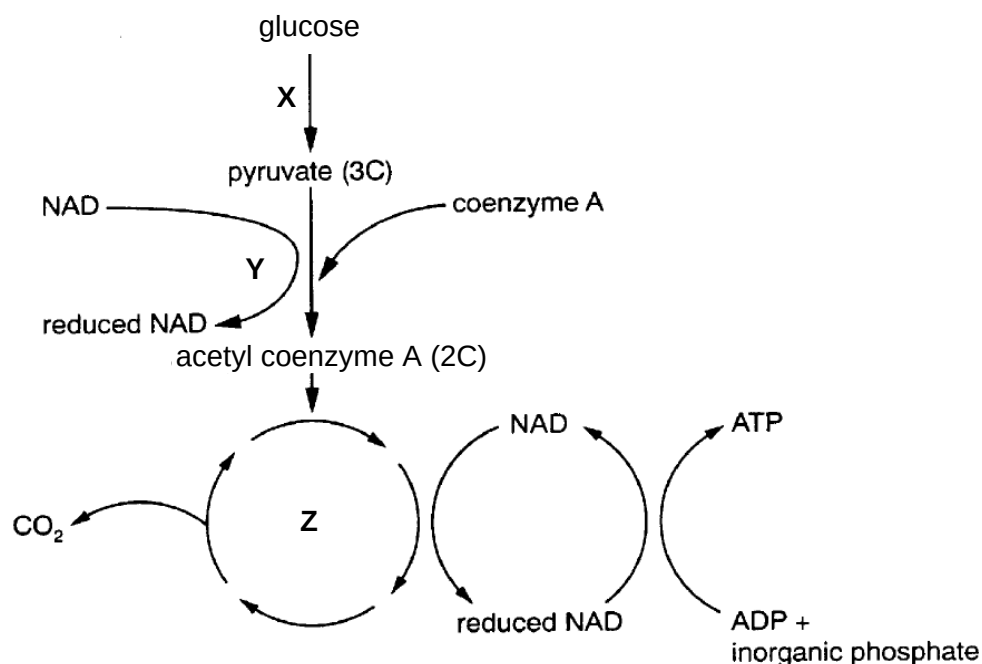


Which of the following can be concluded based on the results?

I	Upon the addition of oxygen, glycolysis and subsequently, link reaction, Krebs cycle and oxidative phosphorylation occurred.
II	Electron transfer was initiated by the addition of oxygen.
III	The pH drop was greater with NADH than with FADH_2 , which is consistent with the greater ATP yield that accompanies the oxidation of NADH.
IV	The rapid decline in pH indicates that protons were pumped into the intermembrane space when oxygen was available.

- A I only
 B II and IV only
 C II, III and IV only
 D All of the above

- 26 The flow chart shows a series of reactions occurring in an animal cell.

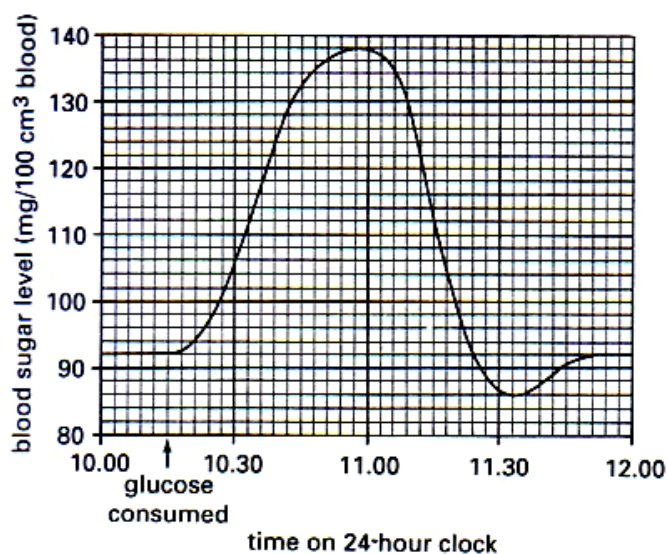


Which of the following statements correctly describes the flow chart?

- A Reaction X, which occurs in the cytosol, is an anabolic reaction.
- B Reaction Y involves the process of substrate-level phosphorylation, whereby pyruvate is first converted to a compound called acetyl coenzyme A.
- C Reaction Y occurs in the cytoplasm whereas reaction Z occurs in the mitochondria.
- D Reaction Z is a catabolic pathway which occurs twice for every glucose molecule to be completely oxidised.
- 27 The cell membrane of an African desert rat and that of the Antarctic penguin are compared to investigate their composition. Which of the following is the most probable makeup of the respective membranes?

	Amount of unsaturated fatty acids		Amount of cholesterol	
	Penguin	Desert rat	Penguin	Desert rat
A	Low	High	Low	High
B	High	Low	Low	Low
C	High	High	High	High
D	High	Low	High	High

- 28 Which of the following statements is **not** true about transport across membranes?
- A The steeper the concentration gradient, the greater the rate of diffusion.
 - B Active transport is the facilitated movement of molecules against its concentration gradient, and requires energy in the form of ATP.
 - C Facilitated diffusion is a form of passive transport that does not require energy.
 - D In an animal cell, osmosis is the net movement of water molecules from a region of less negative solute potential to a region of more negative solute potential.
- 29 The graph shows the blood sugar level of an individual who has consumed 50 g of glucose at the time indicated by the arrow.



Compared with normal levels, extra insulin and extra glucagon would be present at certain times during this two-hour period. Two of these times are

	Extra insulin	Extra glucagon
A	10.30	11.30
B	11.00	12.00
C	10.30	11.15
D	11.45	11.15

- 30 The table shows the velocity of impulse propagation in the nerve fibres of some animals.

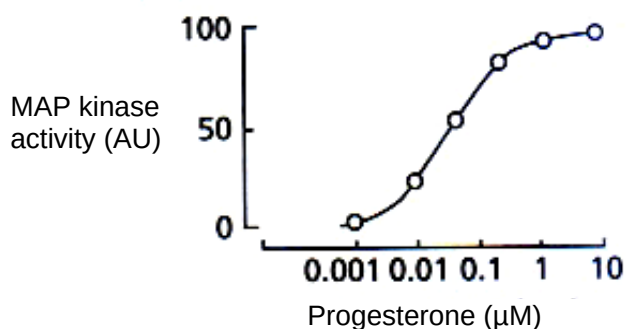
ANIMAL	TEMP (°C)	DIAMETER OF FIBRE (μm)	IMPULSE VELOCITY (m sec^{-1})
Crab nerve fibre	20	30	5
Squid giant axon	20	500	25
Prawn nerve fibre	20	35	20
Frog nerve fibre (myelinated)	20	16	32
Cat nerve fibre (non-myelinated)	38	15	2.3
Cat nerve fibre (myelinated)	38	20	100

Which of the following conclusions can be made?

- A At 20°C, the larger the fibre diameter, the higher the velocity of impulse transmission.
- B If the temperature is kept constant, myelinated nerve fibres conduct impulses faster than non-myelinated fibres only at 38°C.
- C At both 20°C and 38°C, myelination has a greater impact on impulse velocity than fibre diameter.
- D Temperature does not seem to have an impact on impulse velocity.

- 31** Maturation of frog oocytes (fertilised eggs) results from a series of cell signalling events triggered by the hormone progesterone. Progesterone directly stimulates the translation of the mRNA encoding Mos, a protein which sets off a downstream signalling cascade leading to the activation of an enzyme called MAP kinase. MAP kinase directly stimulates oocyte maturation.

In an investigation, 16 frog oocytes were placed in each of six petri-dishes that are treated with six different concentrations of progesterone. The activity level of active MAP kinase in arbitrary units (AU) was measured by the proportion of oocytes that have matured. The results are shown in the graph.



Which of the following can least likely be concluded?

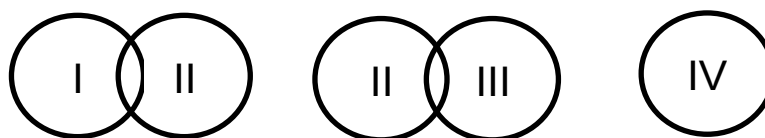
I	Progesterone is a lipid-soluble hormone.
II	Eight oocytes would have matured in the dish with 0.1 μM progesterone.
III	The maturation of frog oocytes is activated by phosphorylation.
IV	The rate of oocyte maturation is highest at 0.1 μM progesterone.
V	Mos is a second messenger.
VI	Transduction is multistep.

- A** II and IV only
B I, III and VI only
C II, IV and V only
D III, V and VI only
- 32** In 1954, copper waste in the Finniss River killed numerous fish. This caused various species in the area to die out. However, one species, the black-banded rainbow fish, increased in numbers. The black-banded rainbow fish have modified gills that enable them to filter and remove copper before it enters their bodies.

With respect to the black-banded rainbow fish, it is reasonable to conclude that in 1954,

- A** a mutation occurred in their population that year.
B the ability of their gills to remove copper already existed.
C the high level of copper in the water changed the structure of their modified gills.
D their genomes are identical to those of the other species of fish that existed.

- 33** The numbers **I**, **II**, **III** and **IV** in the diagram refer to four populations of frogs. These populations are represented diagrammatically by circles. Overlapping circles indicate interbreeding of the populations concerned. Where circles do not overlap, no interbreeding of the populations occurs.



Which of the following correctly reflects the biological concept of species?

	Number of species of frogs	Reasons
A	Two	There is potential gene flow between I and II as well as II and III . Gene flow is possible between I and III . IV is reproductively isolated from the rest.
B	Two	There is potential gene flow between I and II as well as II and III . Gene flow is not possible between I and III . IV is reproductively isolated from the rest.
C	Three	There is potential gene flow between I and II as well as II and III . Gene flow is possible between I and III . IV is reproductively isolated from the rest.
D	Three	There is potential gene flow between I and II as well as II and III . Gene flow is not possible between I and III . IV is reproductively isolated from the rest.

- 34** Which taxa do *Zerynthia rumina* and *Zerynthia polyxena* share?

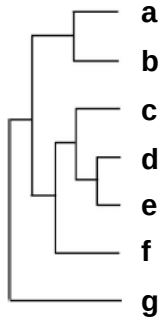
- A** They share the same class but not the same family.
- B** They share the same species but not the same class.
- C** They share the same class but not the same genus.
- D** They share the same family but not the same species.

- 35 The table shows the number of estimated nucleotide substitutions that have occurred since the divergence of seven species **a** to **g**.

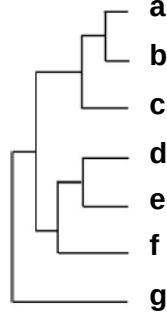
	b	c	d	e	f	g
a	39	72	128	126	159	269
b		81	130	128	158	268
c			129	127	157	267
d				56	154	271
e					151	268
f						273

Which of the following phylogenetic trees best shows the relationship among these seven species?

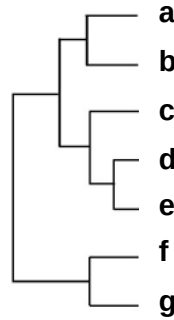
A



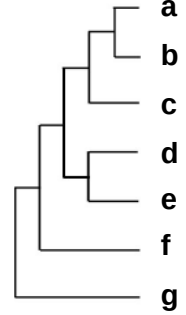
B



C



D



- 36 The table shows the functional properties of three thermostable DNA polymerases (Pol) used in the Polymerase Chain Reaction (PCR).

Enzyme	Relative efficiency ^a	Error rate ^b	Processivity ^c	Extension rate ^d	3' to 5' exonuclease activity	5' to 3' exonuclease activity
<i>Taq</i> Pol	88	2×10^{-4}	55	75	No	Yes
<i>Tli</i> Pol	70	4×10^{-5}	7	67	Yes	No
<i>Pfu</i> Pol	60	7×10^{-7}	n.d.	n.d.	Yes	No

^a Percent conversion of template to product per cycle

^b frequency of errors per base pairs incorporated

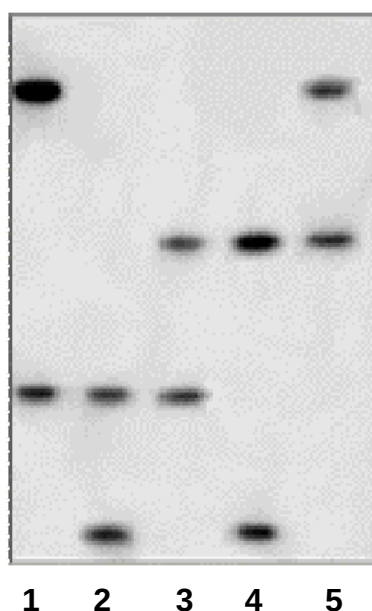
^c Average number of nucleotides added before dissociation

^d Average number of nucleotides added per second

n.d. not determined

Which of the following deductions is **not** true?

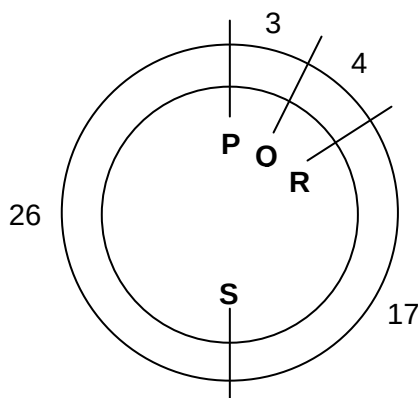
- A *Pfu* Pol is likely to have a lower extension rate than *Tli* Pol.
- B *Taq* Pol has proofreading capability.
- C The lower error rates of *Tli* Pol and *Pfu* Pol can be correlated with their 3' to 5' exonuclease activity.
- D *Taq* Pol has the highest relative efficiency as it has a comparatively high processivity and extension rate.
- 37 Results from the DNA fingerprint analysis of a single VNTR locus for a man and his four different children are shown in the autoradiograph.



Which lane contains the DNA of the father?

- A Lane 1
- B Lane 2
- C Lane 3
- D Lane 4

- 38 The diagram shows a plasmid with the positions of four restriction sites **P** to **S** indicated. The distances between them are shown in kb.



Copies of the plasmid were cut using two different restriction enzymes at a time, and the resulting fragments were separated by gel electrophoresis. The results are shown in the table.

Length of fragment (kb)	Restriction enzymes used		
	<i>BalI</i> and <i>EcoRI</i>	<i>PvuII</i> and <i>AvaI</i>	<i>PvuII</i> and <i>EcoRI</i>
40	■		
30		■	■
20		■	
10			■
	■		

Which of the following correctly matches the four restriction enzymes to their respective restriction sites on the plasmid?

	P	Q	R	S
A	<i>AvaI</i>	<i>EcoRI</i>	<i>PvuII</i>	<i>BalI</i>
B	<i>BalI</i>	<i>AvaI</i>	<i>EcoRI</i>	<i>PvuII</i>
C	<i>PvuII</i>	<i>BalI</i>	<i>AvaI</i>	<i>EcoRI</i>
D	<i>EcoRI</i>	<i>PvuII</i>	<i>BalI</i>	<i>AvaI</i>

- 39** A couple who is planning a pregnancy undergoes genetic screening for cystic fibrosis.

Which of the following correctly describes the characteristic and outcome of the genetic screening?

	Characteristic of the genetic screening	Outcome of the genetic screening
A	Screening of many alleles as there are many possible mutations to the CFTR gene.	High chance of producing a child with cystic fibrosis if both parents are carriers, as it is a recessive condition.
B	Screening of many gene loci as the CFTR gene is part of a gene family.	High chance of producing a child with cystic fibrosis even if one parent is a carrier, as it is a dominant condition.
C	Screening of one allele as mutated CFTR gene is expressed in homozygous individuals.	High chance of producing a child with cystic fibrosis even if one parent is a carrier, as it is a dominant condition.
D	Screening of two alleles as CFTR gene is present in two copies in a diploid organism.	High chance of producing a child with cystic fibrosis if both parents are carriers, as it is a recessive condition.

- 40** Many human proteins, such as growth hormone, are now produced in large quantities by genetically engineered cells. Previously, growth hormone was extracted from animals.

Which of the following is **not** an advantage of producing growth hormone by genetically engineered cells?

- A** It is more cost effective and easier to purify.
- B** There may be fewer side effects as there will be no immune rejection.
- C** The production level can meet demand as there is greater production of the growth hormone.
- D** There is a lower risk of contamination due to transient expression.